

STTPK3123 - Quick Exam Notes

Topic 1: Software Quality Concept & Principle

Key Points:

Software Quality = Meeting requirements + User satisfaction

3 Views of Quality:

1. User-based - Fitness for purpose
2. Developer-based - Conforms to requirements
3. Manufacturing-based - Defect prevention

ISO 25010 Characteristics (8):

1. Functional Suitability ✓
2. Performance Efficiency ⚡
3. Compatibility 🔗
4. Usability 👥
5. Reliability ⌚
6. Security 🔒
7. Maintainability 🔧
8. Portability 📦

McCall's 3 Categories:

1. Product Operation - Correctness, Reliability, Efficiency, Integrity, Usability
2. Product Revision - Maintainability, Flexibility, Testability
3. Product Transition - Portability, Reusability, Interoperability

QA vs QC:

- QA = Process-focused, Prevents defects
- QC = Product-focused, Detects defects

Topic 2: Software Quality Models & Standards

Models Comparison:

Model	Year	Focus	Key Features
McCall	1977	Product	11 factors, 3 categories
Boehm	1978	Product	3 levels: High, Intermediate, Primitive
FURPS	HP	Requirements	F-U-R-P-S (Functionality, Usability, Reliability, Performance, Supportability)

Product-Oriented Approach:

- Focus: Final software product
- When: After development
- Tools: Testing, inspection

Process-Oriented Approach:

- Focus: Development process
- When: Throughout lifecycle
- Tools: Standards, audits, CMMI

Topic 3: Software Process Improvement

CMMI Levels (5):

1. Initial - Ad hoc, chaotic
2. Repeatable - Basic project management
3. Defined - Standardized processes
4. Managed - Measured & controlled
5. Optimizing - Continuous improvement

ISO 15504 (SPICE) Levels (0-5):

1. Incomplete
2. Performed
3. Managed
4. Established
5. Predictable
6. Optimizing

Key Difference:

- CMMI = Model framework
- ISO 15504 = International standard

Topic 4: Software Quality Assurance (SQA)

SQA Plan Contents:

1. Quality objectives
2. Standards & procedures
3. Roles & responsibilities
4. Review/audit schedules
5. Metrics & reporting

Audit Types:

- Process Audit - Checks SDLC compliance
- Product Audit - Evaluates final software
- Internal/External - Who performs it

Audit Steps:

1. Planning
2. Preparation
3. Execution
4. Reporting
5. Follow-up

Non-Conformance Types:

- Major NC - Serious failure
- Minor NC - Small issues
- Observation - Improvement opportunity

Topic 5: Software Quality Control (QC)

QC = Detection Activities:

- Testing
- Inspections
- Reviews
- Product-focused (not process)

Remember: QC finds bugs, QA prevents bugs!

ISO Standards Comparison

ISO 15504 vs ISO 25010:

Aspect	ISO 15504	ISO 25010
Focus	Process quality	Product quality
Measures	Process capability	Software characteristics
Levels	0-5 capability levels	8 quality characteristics
Used by	Organizations, auditors	Developers, testers

Topic 6: Quality Management System (QMS)

QMS Elements:

1. Quality Planning - Set objectives
2. Quality Assurance - Prevent defects
3. Quality Control - Detect defects
4. Standards & Procedures - Guidelines
5. Configuration Management - Version control
6. Documentation - Records
7. Training - Team competency
8. Continuous Improvement - Kaizen

Key Metrics for QMS:

- Defect density
- Defect Removal Efficiency (DRE)
- MTTF/MTTR
- Code coverage
- Review effectiveness

Topic 7: Software Metrics

3 Types of Metrics:

1. Product Metrics - Software quality (defect density, complexity)
2. Process Metrics - Development effectiveness (defect arrival rate)
3. Project Metrics - Progress & productivity (effort, schedule)

Important Metrics:

- Defect Density = Defects / KLOC
- Code Coverage = % code tested
- Cyclomatic Complexity = Code complexity measure
- MTTF = Mean Time To Failure
- MTTR = Mean Time To Repair

CI/CD Metrics:

1. Build Success Rate
2. Deployment Frequency
3. Lead Time for Changes
4. Change Failure Rate
5. MTTR (Mean Time to Recovery)

Topic 8: Review & Inspections

4 Review Types:

Type	Formality	Leader	Focus
Peer Review	Informal	Author/Peer	Feedback + defects
Walkthrough	Semi-formal	Author	Understanding + defects
Technical Review	Medium	Tech Lead	Technical correctness
Inspection	Very formal	Moderator	Defects only

Inspection Roles:

1. Moderator - Leads meeting
2. Author - Created work
3. Reader - Reads material
4. Recorder - Documents defects
5. Reviewers - Experts

Benefits of Reviews:

- Early defect detection
- Knowledge sharing
- Improved quality
- Reduced rework cost

Topic 9: SQA in SDLC

SQA Activities by Phase:

1. Requirements - Reviews, validation
2. Design - Architecture reviews
3. Coding - Code reviews, standards
4. Testing - Test planning, coverage
5. Deployment - Release readiness
6. Maintenance - Change control

Agile SQA Practices:

- Definition of Done (DoD)
- CI/CD pipelines
- Automated testing
- Pair programming
- Sprint reviews
- TDD (Test-Driven Development)

YourNote Summaries

Software Quality Characteristics (ISO 25010):

FPCUR-SMP:

- F - Functional Suitability
- P - Performance Efficiency
- C - Compatibility
- U - Usability
- R - Reliability
- S - Security
- M - Maintainability
- P - Portability

McCall Model (11 factors):

Operation: Correctness, Reliability, Efficiency, Integrity, Usability

Revision: Maintainability, Flexibility, Testability

Transition: Portability, Reusability, Interoperability

Rayleigh Model:

- Bell curve shape
- Predicts effort/defects over time
- Peak in middle of project
- Used for planning/testing

Software Audit vs Testing:

Audit

Testing

Process & compliance

Product quality

Independent

Part of development

Standards & regulations

Functional behavior

Reliability Growth Models:

- Predict reliability improvement
- Examples: Musa-Okumoto, Goel-Okumoto
- Measures: MTTF, failure rate, defect trends

Quick Comparison Tables

Quality Approaches:

Product-Oriented	Process-Oriented
Output-focused	Activity-focused
Reactive	Preventive
After development	Throughout lifecycle
Testing, inspection	Standards, audits

Review Types:

	Peer Review	Walkthrough	Inspection
Formality	Low	Medium	High
Speed	Fast	Medium	Slow
Documentation	Optional	Some	Formal report

ISO Standards:

	ISO 15504	ISO 25010
Also called	SPICE	Quality Model
Measures	Process capability	Product quality
Output	Level 0-5	8 characteristics

Exam Tips

Must-Know:

1. QA vs QC - Process vs Product
2. McCall's 3 categories - Operation, Revision, Transition
3. ISO 25010 - 8 characteristics
4. CMMI levels - 1 to 5
5. Review types - 4 types with differences
6. Metrics - Defect density, MTTF, code coverage
7. Audit types - Process vs Product

Common Questions:

- Difference between product/process approaches
- Which review type when?
- ISO 15504 vs ISO 25010
- Rayleigh curve explanation
- SQA activities in each SDLC phase

Formulas to Remember:

- Defect Density = Defects / KLOC
- DRE = Defects removed / (Defects removed + Defects found later)
- Code Coverage = (Lines tested / Total lines) × 100%

Last-Minute Memory Aids

- **McCall's 11 Factors:**
"CREIU" for Operation (Correctness, Reliability, Efficiency, Integrity, Usability)
"MFT" for Revision (Maintainability, Flexibility, Testability)
"PRI" for Transition (Portability, Reusability, Interoperability)
- **ISO 25010:**
"For Perfect Quality, Use Reliable Software, Maintain Properly"
F-P-Q-U-R-S-M-P
- **CMMI Levels:**
"I Really Do Manage Optimally"
Initial, Repeatable, Defined, Managed, Optimizing
- **Review Types:**
"Peer Walks Technically Inspecting"
Peer Review, Walkthrough, Technical Review, Inspection

MORE INFO

Class	ISO Software Engineering Standards
Product Standards	ISO 9126: Software Product Quality Measurement. ISO 14598: Software Product Evaluation. ISO 25051: Requirements for Quality of Commercial Off-The-Shelf (COTS) Software Product and Instructions for Testing. ISO 15026: System and Software Integrity Levels. ISO 15910: Software User Documentation Process.
Process Standards	ISO 15504: Software Process Assessment. ISO 15939: Software Measurement Process. ISO 6592: Guidelines for the Documentation of Computer Based Application Systems ISO 18019: Guidelines for the Design and Preparation of Software User Documentation. ISO 14102: Guidelines for the Evaluation and Selection of CASE Tools.